PENGXIAO YANG

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EDUCATION

IMPERIAL COLLEGE LONDON MSc Control and Optimisation 2025 - Present

Core Modules: Control Engineering, Optimisation, Modelling and Control of Multi-Body Mechanical Systems, Stability and Control of Non-linear Systems, Discrete-Event Systems, Computer Vision and Pattern Recognition, and Predictive Control.

DALIAN UNIVERSITY OF TECHNOLOGY Bachelor of Science in Engineering 2020 - 2024

Mechanical Design Manufacturing and Its Automation **GPA: 90.0/100**

English proficiency: IELTS Academic Overall **7.5** (L/R/W/S: **8.5/8.5/6.0/6.0**) 06/2025

RESEARCH EXPERIENCE

ALBA: A Soft Robotic, Hoberman Inspired Implant for MRI-Compatible Bladder Voiding Assistance 12/2025-Present

(Imperial College Individual Research Project)

Supervisor: Professor Eric M. Yeatman and Dr Ranan Dasgupta

- Developing and experimentally evaluating a metal-free, Hoberman-inspired auxetic mechanism designed to apply controlled external compression for assisted bladder voiding while remaining compatible with MRI environments.
- Designed and assembled a sensorised bladder-phantom test platform integrating an ESP32, stepper motor, quadrature encoder, FSR-based force sensing, flowmeter and solenoid valve; calibrated the force sensor and implemented real-time sensor fusion combining encoder-derived position, force and flow measurements.
- Established a CAD–FEA–prototype–phantom workflow to relate actuator motion and mechanical loading to bladder deformation, model-estimated pressure and measured urinary flow, supporting energy analysis and system identification; flow-feedback control based on the fused sensor data is under development.
- Prepared a one-page research abstract accepted for IEEE EMBC 2026 and delivered the corresponding poster presentation (Poster No. 3012) in Toronto, Canada.

Fabrication, Modelling and Applications of Ni/Ni_2Al_3 Laminated Intermetallic Composites 10/2024-10/2025

Supervisor: Professor Kailun Zheng and Professor Jingwei Xian

- Fabricated Ni/Ni_2Al_3 laminated intermetallic composites and optimised the processing parameters; prepared specimens and conducted high-temperature tensile tests to characterise their temperature- and strain-rate-dependent superplastic behaviour.
- Derived a microstructure-informed, physically based dual-phase constitutive model; calibrated and validated the model against experimental data and applied it to numerical simulations of thermal deformation and forming behaviour.
- Presented the findings at ICLMM 2025; the resulting first-author article was published in the International Journal of Lightweight Materials and Manufacture, with artwork from the study selected for the journal issue cover. A related comprehensive review article is also in preparation.

Research on the Constitutive Model of AA6011 Under Hot Forming Conditions 01/2024-09/2024

Supervisor: Professor Kailun Zheng

- Conducted high-temperature tensile tests and processed temperature- and strain-rate-dependent material data in MATLAB; derived and calibrated the constitutive equations against experimental results.
- Implemented the calibrated material model as a Fortran VUMAT in Abaqus, establishing an experiment–model–simulation workflow to predict the material response under hot-forming conditions.
- The resulting work contributed to an oral conference presentation and resulted in a published article.

CONFERENCE PRESENTATIONS

- **48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2026), Toronto, Canada, 26-30 July 2026. (**One-page abstract and poster presentation**)**
  Topic: "A Soft Robotic, Hoberman Inspired Implant for MRI-Compatible Bladder Voiding Assistance."
  Poster No. 3012.
- **4th International Conference on Lightweight Materials and Manufacture (ICLMM 2025), Dalian, China, 18-21 April 2025. (**Oral presentation**)**
  Topic: "Synthesis and Modelling of Ni/Al Laminated Intermetallic Composites: Growth Kinetics and

Constitutive Behaviour."

- **International Automotive Lightweight Conference & Exhibition, Yangzhou, China, 9-11 October 2024. (**Oral presentation**)**
  Topic: "Multiscale Modelling and Simulation of Hot Gas Pressure Forming for an Integrated Aluminium-Alloy Longitudinal Beam in a Novel Off-Road Vehicle."

PUBLICATIONS

- **Thermal Deformation Behaviour and Physically-based Dual-phase Constitutive Model of Ni/Ni_2Al_3 Laminated Composites.** (Artwork from this study was selected for the journal issue cover.)
  Journal: *International Journal of Lightweight Materials and Manufacture*
  Co-Authors: **Pengxiao Yang**, Mengmeng Li, Yaping Wang, Bao Wang, Jingwei Xian, Peng Lin, Kailun Zheng
  Publication date: January, 2026 DOI: https://doi.org/10.1016/j.ijlmm.2026.01.002
- **Multi-step Hot Metal Gas Forming of Irregular Tubular Components: Physically Based Simulation and Experimental Validation.**
  Journal: *Journal of Materials Processing Technology*
  Co-Authors: Zhihan Wang, **Pengxiao Yang**, Xinyuan Gao, Zhennan Bao, Zhubin He, Kailun Zheng
  Publication date: March, 2026 DOI: https://doi.org/10.1016/j.jmatprotec.2026.119213
- **Microstructure Evolution and Properties of Ti-Nb-Zr-O High Elastic Titanium Alloy during Self-resistance Heating Treatment Process.**
  Journal: *Aerospace Manufacturing Technology*
  Co-authors: Zhihan Wang, **Pengxiao Yang**, Dechong Li, Kun Zhang, Yujie Han, Kailun Zheng
- **Analysis of Vibration Attenuation Characteristics of Large-thickness Carbon Fiber Composite Laminates.**
  Journal: *Viser Technology*
  Co-authors: Yi-Qi Wang, Chaoqun Wang, **Pengxiao Yang**, Ziao Wang, Tete Cao
  Publication date: May, 2022 DOI: https://doi.org/10.33142/mes.v4i1.7514

INDUSTRY AND ENGINEERING EXPERIENCE

Wonwell Autoparts (Dalian) Co., Ltd Engineering Intern 07/2023-08/2023

Conducted material characterisation, finite-element simulations and forming trials in collaboration with factory engineers, contributing to the component's transition to mass production with an annual capacity of 2,000 parts.

Hot Metal Gas Forming of Complex Alloy Components Team Leader 07/2023-08/2023

Performed Abaqus FEA to optimise hot-gas-forming process parameters, manufactured complex aluminium-alloy components through laboratory forming trials, and validated the simulated forming behaviour through mechanical testing; subsequently supported the transfer of the validated process to industrial production.

Modular Quadcopter and Payload System Team Leader 07/2023-08/2023

Led the design and fabrication of a modular interlocking airframe and payload structure designed for rapid assembly, simplified maintenance and replacement of damaged components; optimised the lightweight 3D-printed parts, integrated Arduino-based colour recognition and distance sensing, and tuned the PID controller, contributing to the team's first-place finish in the module competition.

COMPETITIONS

Interdisciplinary Contest in Modeling (ICM), COMAP | Finalist 2023
China Mechanical Engineering Innovation and Creativity Competition | Third Prize 2022

AWARDS

- Dalian University of Technology 2024 Outstanding Graduates 2024
- Scholarship | Chiang Chen Scholarship 2024-2025
- Scholarship | The Learning Excellence Award 2020-2021, 2021-2022, 2022-2023
- Scholarship | The Weichai Power Scholarship 2020-2021
- Scholarship | The Best Social Practice Prize 2020-2021

TECHNICAL SKILLS

MATLAB, Python, Abaqus, COMSOL Multiphysics, SolidWorks, Fusion 360, LaTeX and Microsoft Office.